



INTRODUCTION TO AI & ML FUNDAMENTALS



WHAT IS AI?

01

- AI stands for Artificial Intelligence.
- AI enables machines to think and make decisions like humans.
- AI can learn, solve problems, recognize speech, and understand images.

02

Examples:

- ChatGPT
- Siri
- Alexa
- Self-driving cars



WHAT IS ML?

01

- ML stands for Machine Learning.
- ML is a subset of AI.
- Machines learn from data instead of being explicitly programmed.

02

Examples:

- Netflix recommendations
 - Spam email filtering
 - YouTube suggestions
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AI VS ML

AI

- Broad concept
- Mimics human intelligence
- Focuses on decision-making
- Example: Robotics

ML

- Subset of AI
- Learns from data
- Focuses on predictions
- Example: Recommendation systems

TYPES OF AI



Narrow AI

- Designed for specific tasks
- Most common AI today

Examples:

- Google Assistant
- Face recognition



General AI

- Human-level intelligence
- Can perform multiple tasks like humans
- Still under research

TYPES OF ML



Supervised Learning

- Uses labeled data
- Predicts outputs

Examples:

- House price prediction
- Email spam detection



Unsupervised Learning

- Uses unlabeled data
- Finds hidden patterns

Examples:

- Customer segmentation
- Clustering



Reinforcement Learning

- Learns using rewards and punishments

Examples:

- Robotics
- Game AI

APPLICATIONS OF AI/ML

Healthcare

- Disease prediction
- Medical diagnosis

Finance

- Fraud detection
- Credit scoring

Education

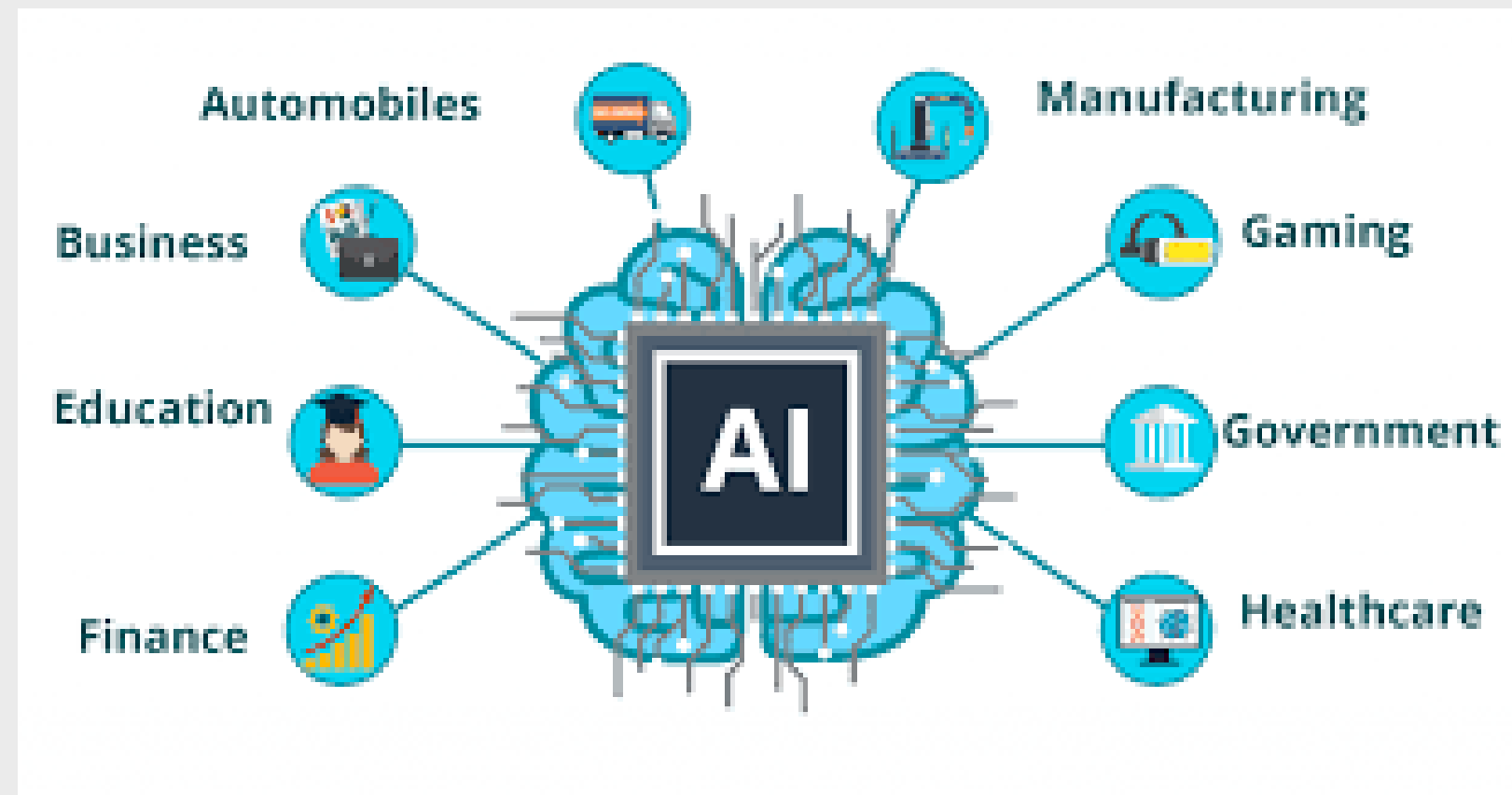
- Personalized learning

Transportation

- Self-driving cars
- Traffic prediction

E-commerce

- Product recommendations
- Chatbots





AI DEVELOPMENT PROCESS

Problem Identification



Data Collection



Data Cleaning



Model Training



Testing



Deployment





CONCLUSION

- AI and ML are transforming industries.
 - They improve automation and decision-making.
 - Understanding AI/ML fundamentals is important for future technologies and careers.
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THANK YOU

